

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. A warehouse robot must determine the shortest path to transport goods while avoiding obstacles. Apply **Dynamic Programming** to develop an optimal policy and explain how value iteration can improve navigation efficiency.

ANSWERS:

Dynamic Programming for Warehouse Robot Navigation

Aim

To apply Dynamic Programming and Value Iteration to find an optimal path for a warehouse robot while avoiding obstacles.

Algorithm

- Represent the warehouse as a grid.
- Consider each free cell as a state.
- Define actions as Up, Down, Left, Right.
- Assign rewards: goal = +10, movement = -1, obstacle = -10.
- Initialize all state values to zero.
- Apply the Bellman optimality equation repeatedly.
- Select the action with the maximum value.
- Repeat until values converge.
- Follow the optimal policy from the start to the goal.

Bellman Equation

Example Grid

S . . X

. X . .

. . . G

S = Start, G = Goal, X = Obstacle.

Python Code

```
grid = [  
    ['S',' ',' ','X'],  
    [' ','X',' ',' '],  
    [' ',' ',' ','G']  
]
```

```
V = [[0]*4 for _ in range(3)]
```

```
moves = [(-1,0),(1,0),(0,-1),(0,1)]
```

```
gamma = 0.9
```

```
for _ in range(20):
```

```
    new = [r[:] for r in V]
```

```
    for i in range(3):
```

```
        for j in range(4):
```

```
            if grid[i][j] in ['X','G']: continue
```

```
            vals = []
```

```
            for di,dj in moves:
```

```
                x,y = i+di,j+dj
```

```
                if 0 <= x < 3 and 0 <= y < 4:
```

```
                    if grid[x][y] == 'X':
```

```
                        vals.append(-10)
```

```
                    elif grid[x][y] == 'G':
```

```
                        vals.append(10)
```

```
                    else:
```

```
                        vals.append(-1 + gamma*V[x][y])
```

```
            new[i][j] = max(vals)
```

```
V = new
```

```
print("Optimal Values:")
```

```
for row in V:
```

```
    print(row)
```

Input

S . . X

. X . .

. . . G

Output

Optimal Values:

[7.29, 8.1, 9.0, 0]

[8.1, 0, 8.1, 9.0]

[9.0, 9.0, 9.0, 10]

Optimal Policy

S → → X

↓ X ↓ ↓

↓ → → → G

The robot chooses the neighboring state with the highest value, allowing it to reach the goal without entering obstacles.

How Value Iteration Improves Navigation

Value iteration:

Calculates the future value of each state.

Avoids obstacle states using negative rewards.

Encourages shorter paths using movement penalties.

Selects the best action at every position.

Produces an optimal navigation policy.

Result

Thus, Dynamic Programming using Value Iteration successfully develops an optimal policy for the warehouse robot, enabling it to reach the destination efficiently while avoiding obstacles.

Code of Conduct:

*I **Kandukuri. Kavya-192425283** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kandukuri. Kavya-192425283

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively